

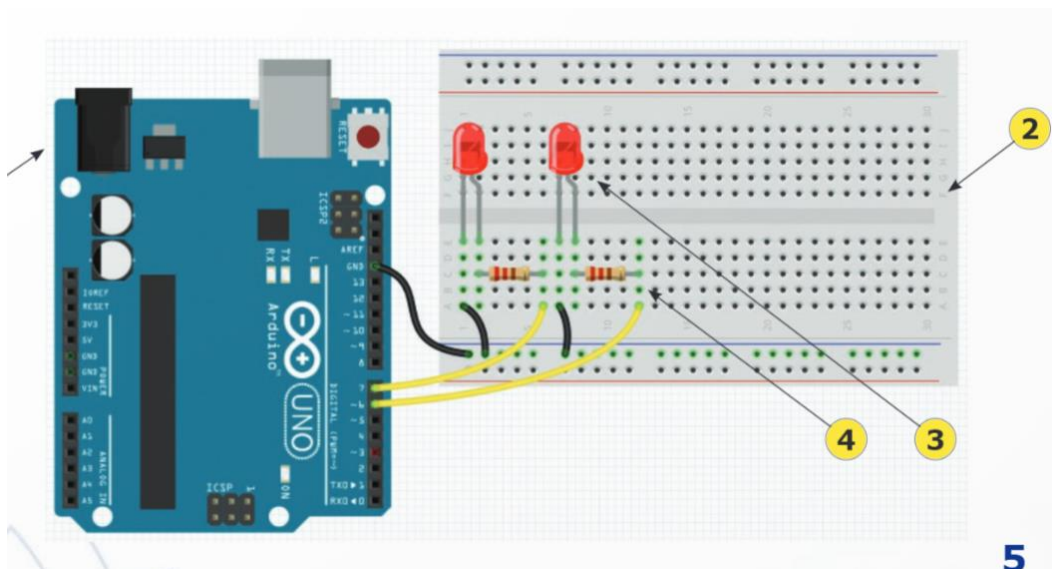


Stage	Subject	Week	Practical Worksheet	Date
G 11	Arduino	7	Mid term Revision	

Student name\----- class\-----

Using Arduino kits and Arduino IDE, design the following circuits

Circuit number 1: Design a circuit that contains two LEDs and turn on and off alternately for 3 seconds



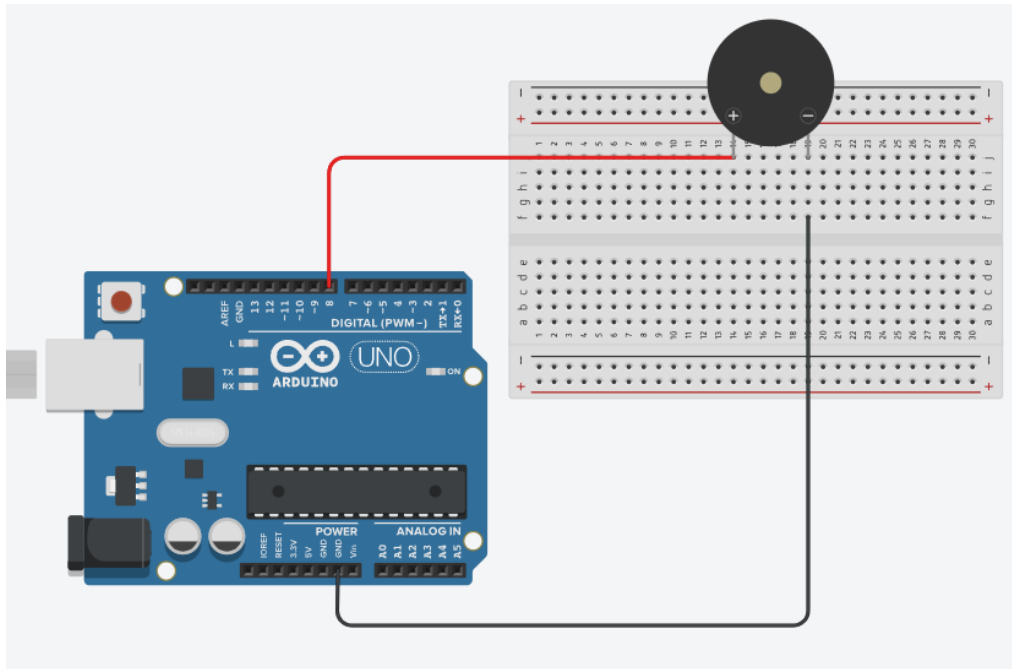
```
int led1=6;
int led2=7;
void setup()
{
  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);
}

void loop()
{
  digitalWrite(led1,HIGH);
  digitalWrite(led2,LOW);
  delay(3000);
  digitalWrite(led1,LOW);
  digitalWrite(led2,HIGH);
  delay(3000);
}
```

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Circuit number 2: Design a circuit that contains a buzzer that produce sounds for 2 seconds and stops for 2 seconds

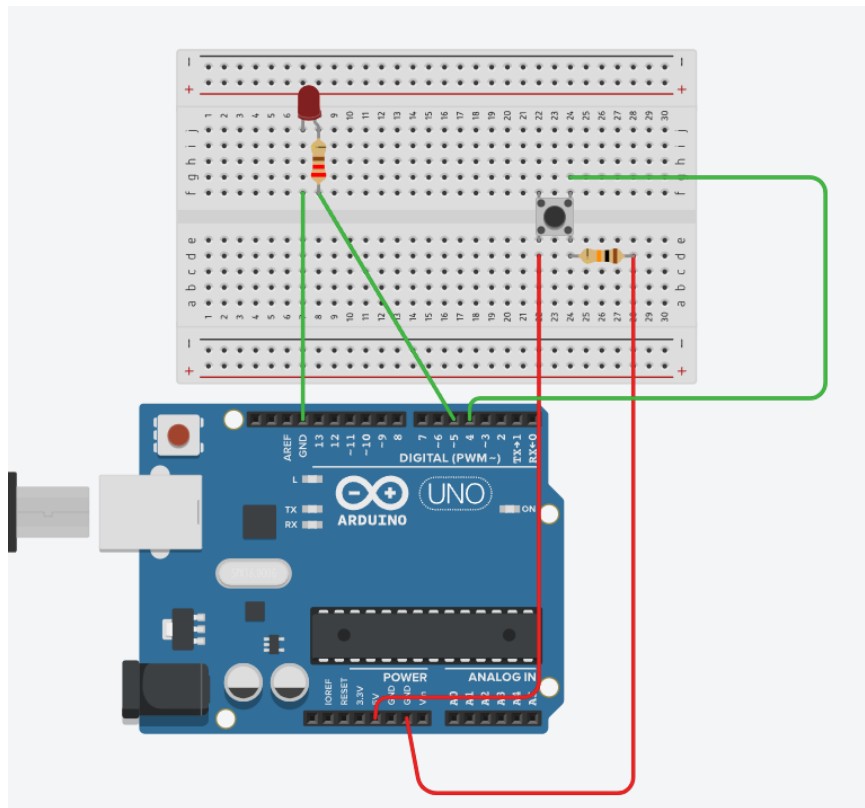


```
int buzzerPin = 8; // اختر أي مخرج رقمي متاح

void setup() {
  pinMode(buzzerPin, OUTPUT);
}

void loop() {
  digitalWrite(buzzerPin , HIGH);
  delay(2000);
  digitalWrite(buzzerPin , LOW);
  delay(2000);
}
```

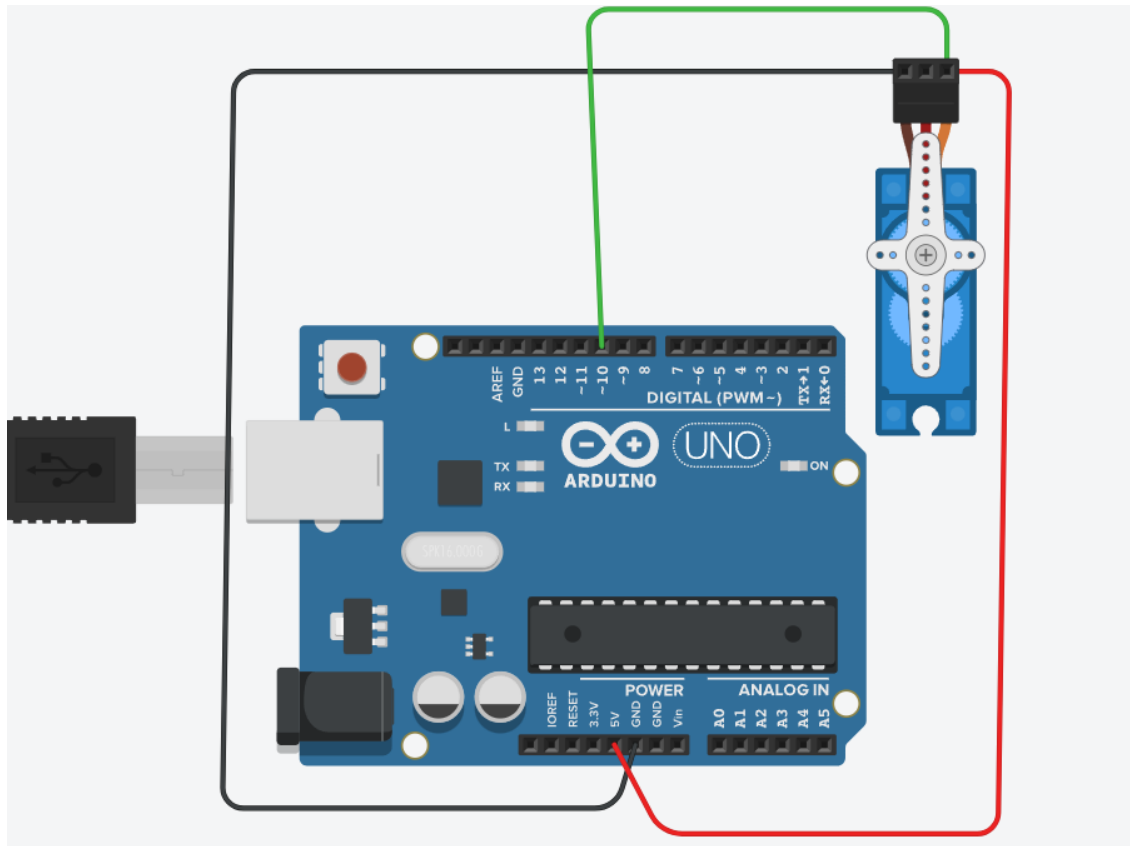
Circuit number 3: Design a circuit that contains a button that turn on a led when pressed



```
// C++ code
//
int led=5;
int buttonpin=4;
int buttonstatus;
void setup()
{
  pinMode(led, OUTPUT);
  pinMode(buttonpin , INPUT);
}

void loop()
{
  buttonstatus=digitalRead(buttonpin);
  if (buttonstatus==HIGH)
  {
    digitalWrite(led , HIGH);
    delay(2000);
  }
  else
  {
    digitalWrite(led , LOW);
  }
}
```

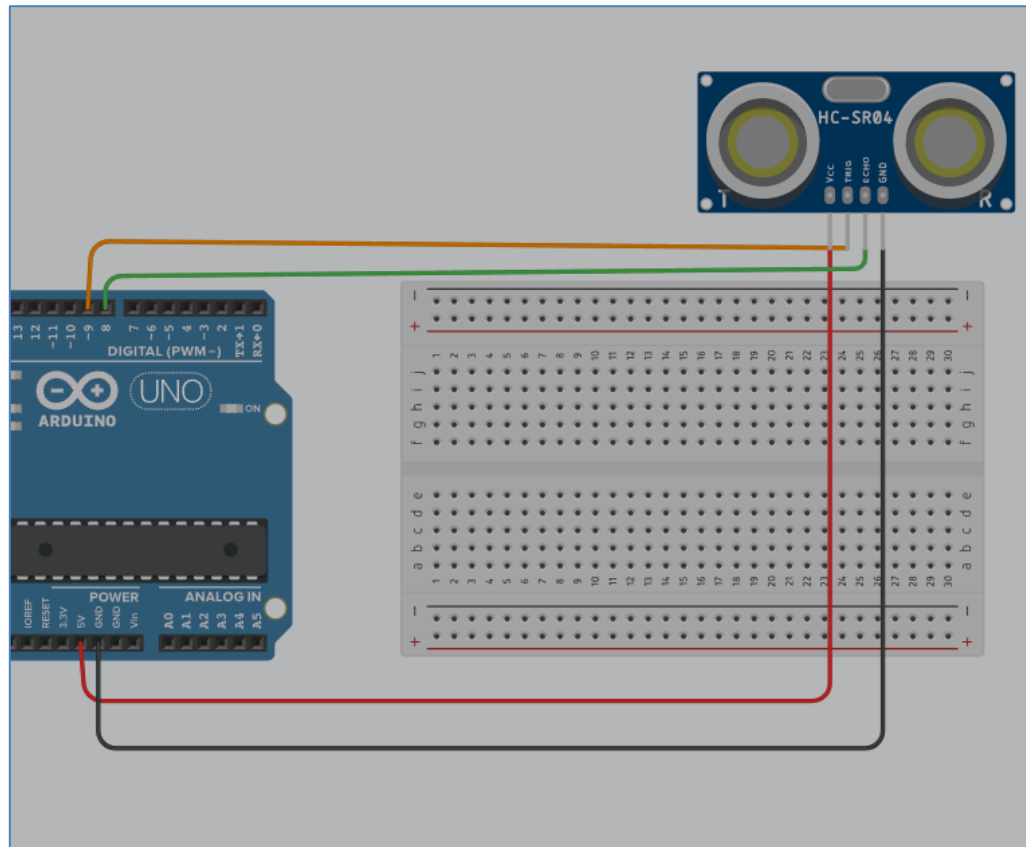
Circuit number 4: Design a circuit that a servo motor that turn 180 degree to the right for 2 seconds and get back to its original position for 2 seconds



```
// C++ code
//
#include<Servo.h>
Servo myservo;
void setup()
{
  myservo.attach(10);
}

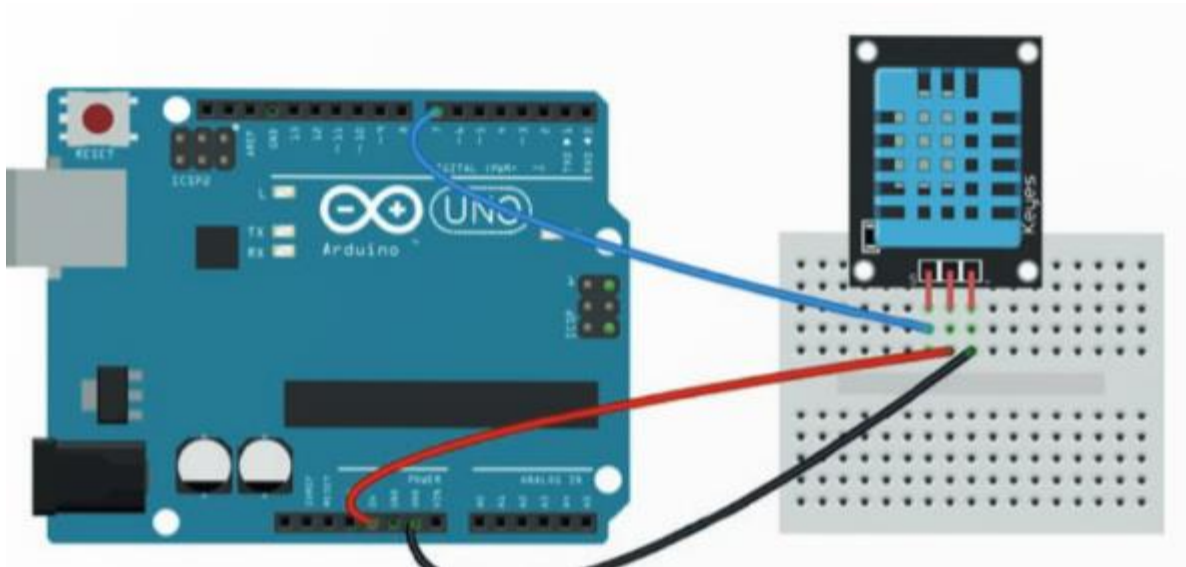
void loop()
{
  myservo.write(180);
  delay(5000);
  myservo.write(0);
  delay(5000);
}
```

Circuit number 5: Design a circuit that contains an Ultrasonic sensor then , display the measured distance on the serial monitor



```
int TRIG_PIN = 9 ;    // الرجل الموصولة بـ Trig
int ECHO_PIN = 8 ;    // الرجل الموصولة بـ Echo
long duration;
int distance;
void setup() {
  Serial.begin(9600);    // تشغيل المراقب التسلسلي
  pinMode(TRIG_PIN, OUTPUT); // كمخرج Trig ضبط
  pinMode(ECHO_PIN, INPUT);  // كمدخل Echo ضبط
}
void loop() {
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);
  // قراءة المدة التي تستغرقها النبضة
  duration = pulseIn(ECHO_PIN, HIGH);
  // حساب المسافة بالسنتيمتر
  distance = duration * 0.034 / 2;
  // طباعة النتيجة
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" CM");
  delay(500);
}
```

Circuit number 6: Design a circuit that contains DHT sensor to measure the temperature and the humidity then , display the values inside the serial monitor



```
#include <dht.h>

#define dht_pin A1
dht DHT;

void setup(){
  Serial.begin(9600);
  delay(300);
  Serial.println("Humidity and temperature\n");
  delay(1000);
}

void loop(){
  → DHT.readll(dht_pin);
  Serial.print("Current humidity = ");
  → Serial.print(DHT.humidity);
  Serial.print("% ");
  Serial.print("temperture = ");
  → Serial.print(DHT.temperature);
  Serial.println("C ");
  delay(5000);
  Serial.println();
}
```